

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			Any three (x1) from: Temperature Time Rate of spin Volume/ mass/ amount of DNA/ sample relevant condition of tube ie {density / concentration / volume / height of/ type of } medium Not solution unqualified 3 correct = 2 marks, 2/1 correct = 1 mark, 0 correct = 0 marks		2		2		2
	(b)	(i)		A: ^{14}N /light/14 B: $^{14}\text{N} + ^{15}\text{N}$ /light and heavy/14 and 15 C: ^{15}N / heavy/ 15 All correct = 1 mark		1		1		1
		(ii)		A. {Generation 0 /{band/peak} at C} is all {heavy isotope/ N^{15} } (1) B. Generation 1 {band/ peak} at B is {a mixture of heavy and light isotope/ an intermediate}(1) C. Because each molecule retains one strand of heavy isotope and one strand of newly formed DNA of light isotope (1) Accept hybrid DNA for intermediate DNA	2			3		
		(iii)		A. (Only) { ^{14}N / light isotope} (nucleotides) are available for replication (1) B. Generation 2 has equal {height scans/ peaks / density} (1) C. because one strand of heavy and one strand of light DNA have been used as templates for the formation of new DNA molecules/ owtte (1) D. Generation 3 has a higher peak of { ^{14}N / light isotope/ A} because more light DNA (strands) are used as template/ owtte (1)		3		4		
							1			